

BambuTagger-AMS

Multi-spool NFC tag reader for Bambu Lab printers. Reads 4 Bambu Lab filament spool tags via RC522, displays live printer AMS tray data over MQTT, and sends RFID tag data to the printer/BMCU. Fully configurable via web interface with automatic AP fallback and OTA firmware updates.

Features

- **4x RC522** on shared SPI bus polling MIFARE Classic 1K + NTAG tags via `MFRC522-spi-i2c-uart-async`
- **Multi-tag auto-detect** – MIFARE Classic 1K (Bambu Lab) or NTAG (SpoolEase), auto-routed to correct parser
- **HKDF-SHA256 key derivation** – derives per-sector MIFARE keys from the tag UID using Bambu's KDF salt
- **Live printer AMS sync** – reads tray data (material, color, type) from the printer over MQTT
- **Bambu BMCU support** – sends `ams_filament_setting` with correct `tray_type`, `tray_color`, `nozzle_temp_min/max`
- **NTAG / SpoolEase** – reads NDEF URI records, extracts spool data from `tag.spoolease.io` URLs
- **Web interface** – 3-tab SPA: Status (merged slots + Sub type + color swatches), Printer Config, WiFi Config
- **OLED display** – boot splash, live AMS tray data, OTA progress bar, MQTT+PTR status
- **4x WS2812 LEDs** – per-slot color display from tag `colorHex`
- **Boot splash**: 2-second logo display from `Logo/splash.png`
- **MQTT bridge** – subscribes to printer status, publishes `ams_filament_setting` commands
- **Auto AP fallback** – captive portal on `192.168.4.1` when WiFi is unavailable
- **OTA updates** – one-click firmware update from GitHub Releases, OLED progress bar, web overlay with status
- **GitHub Actions** – CI build on commit, merged binary + OTA binary release on version tags

Hardware

- **ESP32** Dev Module
- **4x RC522** RFID/NFC readers (SPI, shared bus)
- **4x WS2812** addressable LEDs (daisy-chained, single data pin)
- **128x64 OLED** SSD1306 I2C display

Wiring

Component	ESP32 Pin
SPI MOSI	GPIO 23
SPI MISO	GPIO 19
SPI SCK	GPIO 18
RC522 #1 SS	GPIO 13
RC522 #2 SS	GPIO 12
RC522 #3 SS	GPIO 14
RC522 #4 SS	GPIO 27
RC522 #1 RST	GPIO 26
RC522 #2 RST	GPIO 25
RC522 #3 RST	GPIO 33
RC522 #4 RST	GPIO 32
WS2812 data	GPIO 15
OLED SDA	GPIO 21
OLED SCL	GPIO 22

All RC522 share the same SPI bus (MOSI, MISO, SCK). Each has its own SS and RST pin.

Slot Mapping

RC522 #	SS Pin	Slot	WS2812 Pixel
1	13	1	0
2	12	2	1
3	14	3	2
4	27	4	3

Software Setup (Arduino IDE)

1. Install **ESP32 board package**:
- File → Preferences → Additional Board Manager URLs:

https:// espressif.github.io/arduino-esp32/package_esp32_index.json

Then Tools → Board → Boards Manager → search "ESP32" → install.

2. Install required libraries via **Tools → Manage Libraries**:

- **MFRC522-spi-i2c-uart-async** — multi-reader SPI sharing (not standard MFRC522)
- **Adafruit NeoPixel** by Adafruit (v1.12+)
- **Adafruit GFX Library** by Adafruit (v1.11+)
- **Adafruit SSD1306** by Adafruit (v2.5+)
- **PubSubClient** by Nick O'Leary (v2.8+)
- **ArduinoJson** by Benoit Blanchon (v6.x or v7.x)
- **mbedTLS** — bundled with ESP32 core, used for HKDF-SHA256

3. Open `BambuTagger-AMS.ino`, select **ESP32 Dev Module** as board, and upload.

WiFi & AP Mode

On first boot (or if saved WiFi credentials are invalid), the device automatically opens an access point:

Scenario	Behavior
No WiFi configured	Opens AP immediately
WiFi connection fails	Opens AP after 15 seconds
AP active, credentials exist	Retries STA connection every 30 seconds
STA connects while AP active	Closes AP, switches to normal mode

- **SSID**: Device name (default: `BambuTagger-AMS`)
- **Security**: Open (no password)
- **IP**: `192.168.4.1`
- **Captive portal**: DNS redirects all domains to the config page

Web Interface

Available at `http://<esp32-ip>` on your network, or `http://192.168.4.1` in AP mode.

Status Tab

- **Merged Slot Status** — each slot shows AMS data (Type, Sub, Material, Color) + scanned tag data
- **Color swatches** — 36x36px right-aligned per slot
- **Tag info row** — Tag: `PLA Basic - GFA00` · `C12E1FFF` · `1000g/1000g`

- **Printer AMS Cards** — all detected AMS units with tray grids
- **Update Firmware** — one-click OTA from GitHub Releases

Printer Config Tab

- Printer IP, Port (default 8883), Access Code, Serial Number
- **AMS Unit selector** (A/B/C/D)

WiFi Config Tab

- SSID, Password, Device Name

Footer

- Sticky footer: © 2026 by VID-PRO with link to www.vid-pro.de

OLED Display (128x64)

Device Name		WiFi
1: PLA	#C0C0C0FF	← AMS tray data
2: empty		
3: empty		
4: empty		
MQTT:OK		PTR:OK

OTA progress shown on OLED with header/footer preserved:

"OTA Update" → "Downloading..." → "Flashing... 45%" → auto-reboot

Printer Communication

Subscribe

- **Topic:** `device/<serial>/report`
- **Data:** `push_status` (periodic, ~3KB), `get_version` responses

Publish

- **Topic:** `device/<serial>/request`
- `ams_filament_setting` – structure:

```
{
  "print": {
    "sequence_id": "0",
    "command": "ams_filament_setting",
    "ams_id": 0,
    "tray_id": 0,
    "tray_info_idx": "GFA00",
    "tray_color": "RRGGBBFF",
    "nozzle_temp_min": 190,
    "nozzle_temp_max": 230,
    "tray_type": "PLA"
  }
}
```

- `tray_type` derived from filament index prefix (GFA→PLA, GFG→PETG, etc.)
- `tray_color` forced to RRGGBBFF format
- `nozzle_temp_min/max` from tag block 6

Tag Format & Reading

Bambu Lab (MIFARE Classic 1K)

Bambu Lab uses **MIFARE Classic 1K** tags with fixed block offsets:

Block	Content
0	UID (4 bytes)
1	Variant ID + Material index (e.g. "GFA00")
2	Filament type short name
4	Detailed type string (e.g. "PLA Basic")
5	RGBA color (bytes 0-3) + spool weight LE (bytes 4-5)
6	Nozzle temps (bytes 8-11 LE)

SpoolEase (NTAG)

SpoolEase uses **NTAG** tags with NDEF URI records. The URL contains encoded spool data:

```
https://tag.spoolease.io/S1/?
TG=...&M=PLA&CC=000000FF&SC=GFL99&WL=1000&WE=179&WF=1215&NN=190&NX=240
```

URL parameters parsed and mapped to SpoolInfo:

Param	Field	Description
M=	display type	e.g. "PLA", "PETG"
SC=	materialType	Bambu index for MQTT (e.g. "GFL99")
CC=	colorHex	RGBA hex (e.g. "000000FF")
B=	manufacturer	Brand name (e.g. "Jayo")
WL=	remainingGrams	Remaining filament weight
WE=	empty spool	Empty spool weight
WF=	full spool	totalGrams = WF - WE
NN=	nozzleTempMin	Min nozzle temp °C
NX=	nozzleTempMax	Max nozzle temp °C

Reading uses NDEF TLV parsing:

- NDEF Message TLV (0x03) → NDEF record with TNF=WellKnown, type="U"
- URI identifier code byte prepended to URI string
- Non-printable bytes terminate the URI scan
- Spool ID extracted from URL path after last /

Authentication & Reading

- **Tag auto-detect:** SAK-based type detection (MIFARE 1K vs NTAG)
- **HKDF-SHA256** derives 16 per-sector Key A/B from 4-byte UID
- Bambu KDF salt/info vectors from reverse-engineered firmware
- Falls back to default key 0xFF...FF for blank sectors
- Failed auth re-wakes tag via antenna power-cycle
- Dead readers auto-skipped (version register 0x92/0x91/0xB2 check)
- SPI: 1 MHz via MFRC522_SPI
- NTAG: page-level reads (page+=4, 4 pages per MIFARE_Read)

Filament Type Mapping

Prefix	Type
GFA-GFE, GFL	PLA
GFG	PETG
GFH, GFI	ABS
GFJ	ASA
GFK	TPU

OTA Updates

- **Button:** "Update Firmware" on Status page (shows "Update to vX.Y.Z" or "up to date")
- **Overlay:** full-screen progress overlay with spinner, status text, progress bar
- **Endpoint:** `POST /api/ota` triggers update, `GET /api/ota-check` checks for newer version
- Downloads latest `.ino.bin` from GitHub Releases, flashes via `Update.h`
- 3 retry attempts with 5s stall detection, fresh HTTP client per attempt
- OLED shows "Checking version..." → "Downloading..." → "Flashing..." with percentage
- Device auto-reboots after successful flash, web UI auto-reloads

CI / CD

Workflow at [GHACTIONS/release.yml](#) :

- **On push/PR:** compiles sketch, uploads artifacts
- **On release tag:** creates merged flash binary + OTA binary, attaches to GitHub Release
- Arduino cache for fast rebuilds, pinned esp32:[esp32@3.0.7](#) core

Configuration Defaults

Setting	Default
WiFi SSID	(empty)
WiFi Password	(empty)
Device Name	BambuTagger-AMS
Printer IP	192.168.1.100
Printer Port	8883

Setting	Default
Access Code	(empty)
Printer Serial	(empty)
AMS Unit	A (0)
MQTT Enabled	No
MQTT Update Interval	5000 ms
RFID Poll Interval	100 ms
Firmware Version	1.0.3

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